Introduction to Macroeconomics (S102016CR)

Spring Semester

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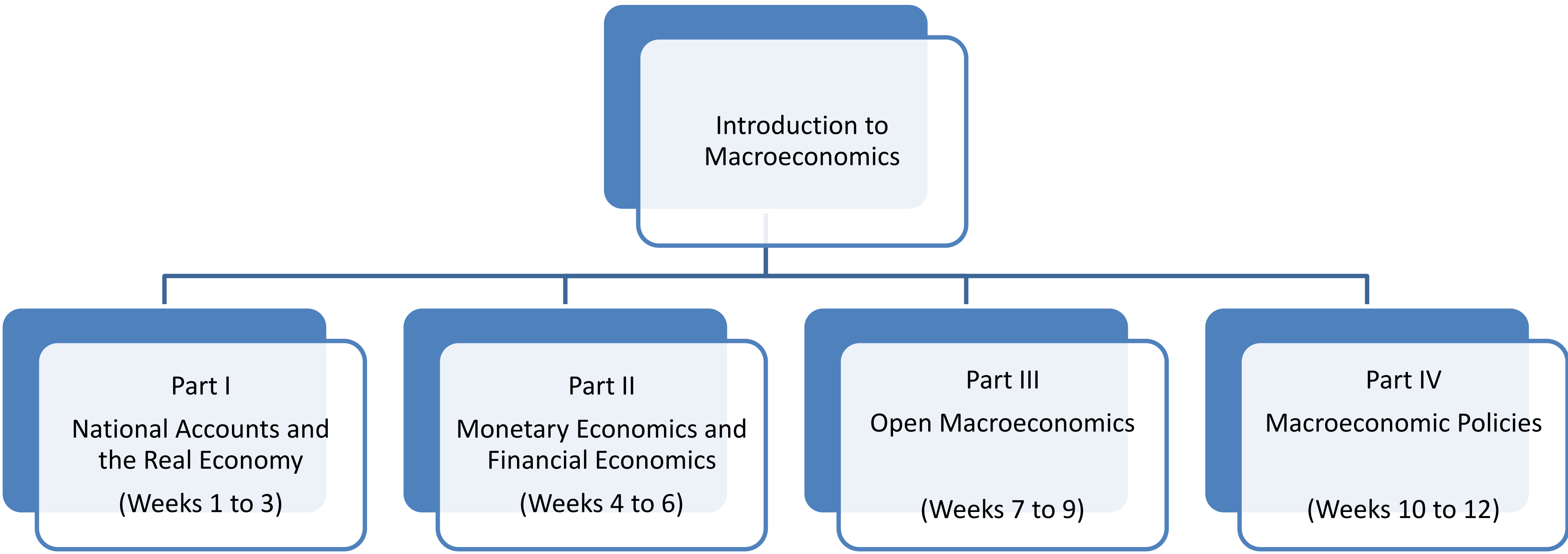
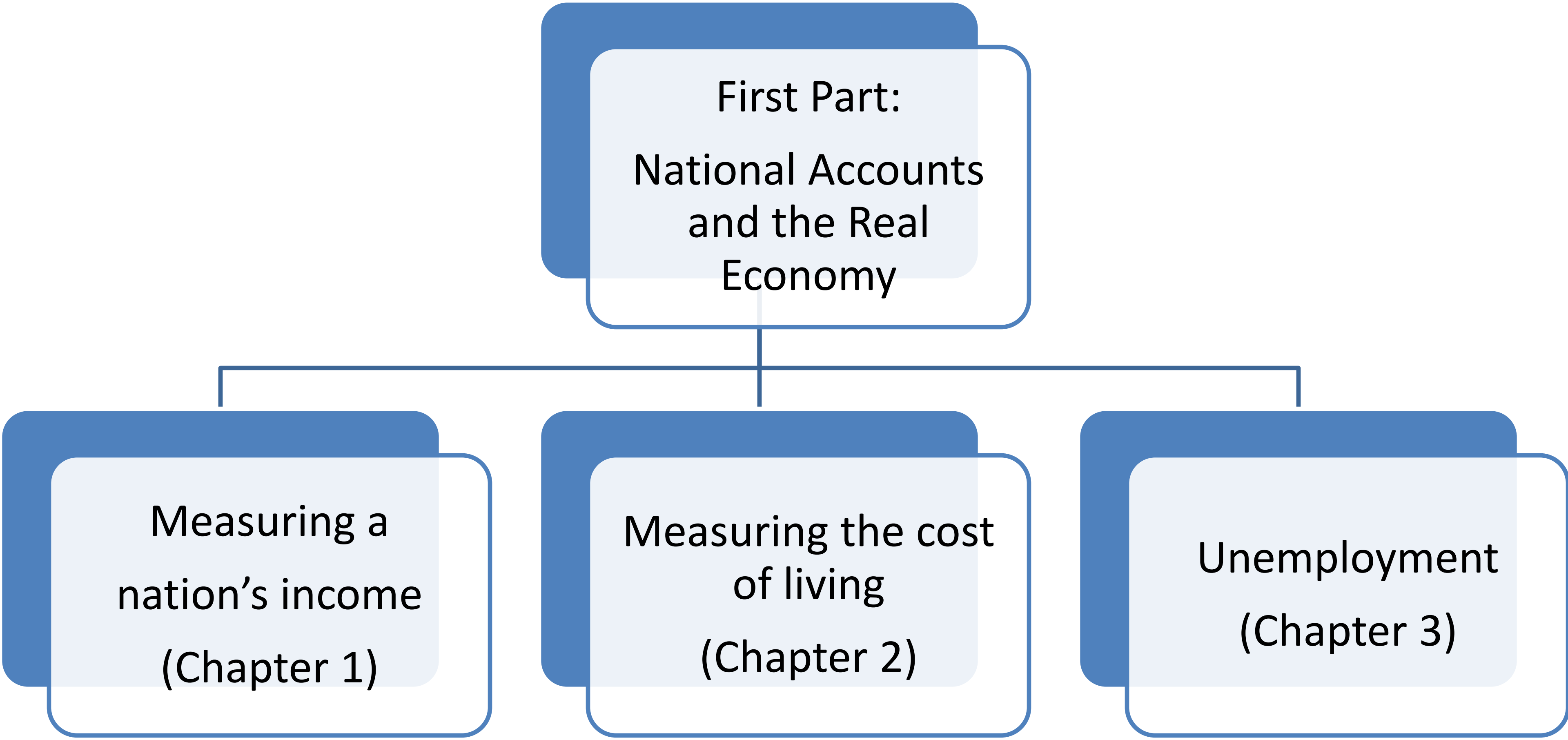
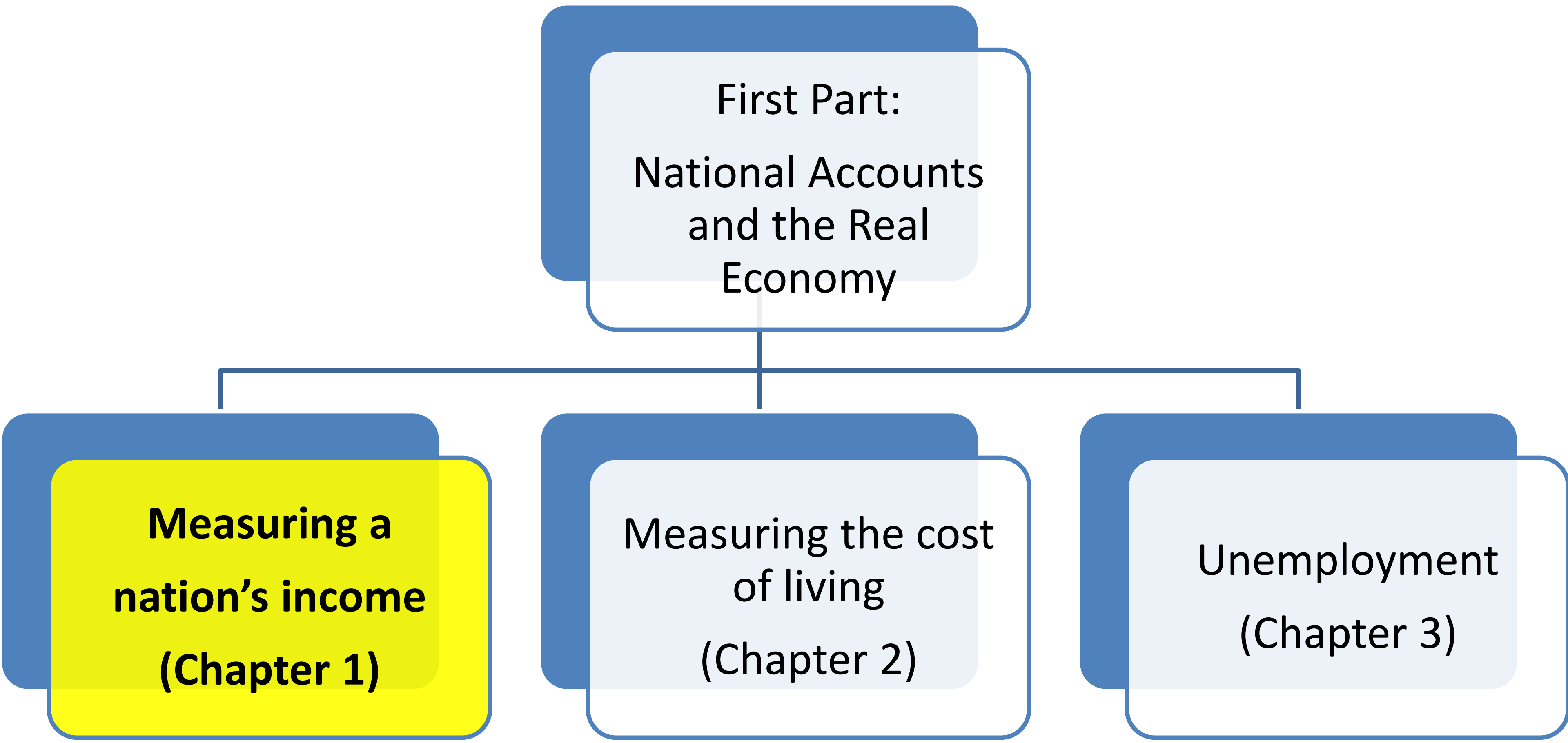


Introduction to Macroeconomics
Part I
National Accounts and the Real Economy
(Weeks 1 to 3)
Part II
Monetary Economics and Financial Economics
(Weeks 4 to 6)
Part III
Open Macroeconomics
(Weeks 7 to 9)
Part IV
Macroeconomic Policies
(Weeks 10 to 12)


First Part:
National Accounts and the Real Economy
Measuring a nation's income
(Chapter 1)
Measuring the cost of living
(Chapter 2)
Unemployment
(Chapter 3)


First Part:
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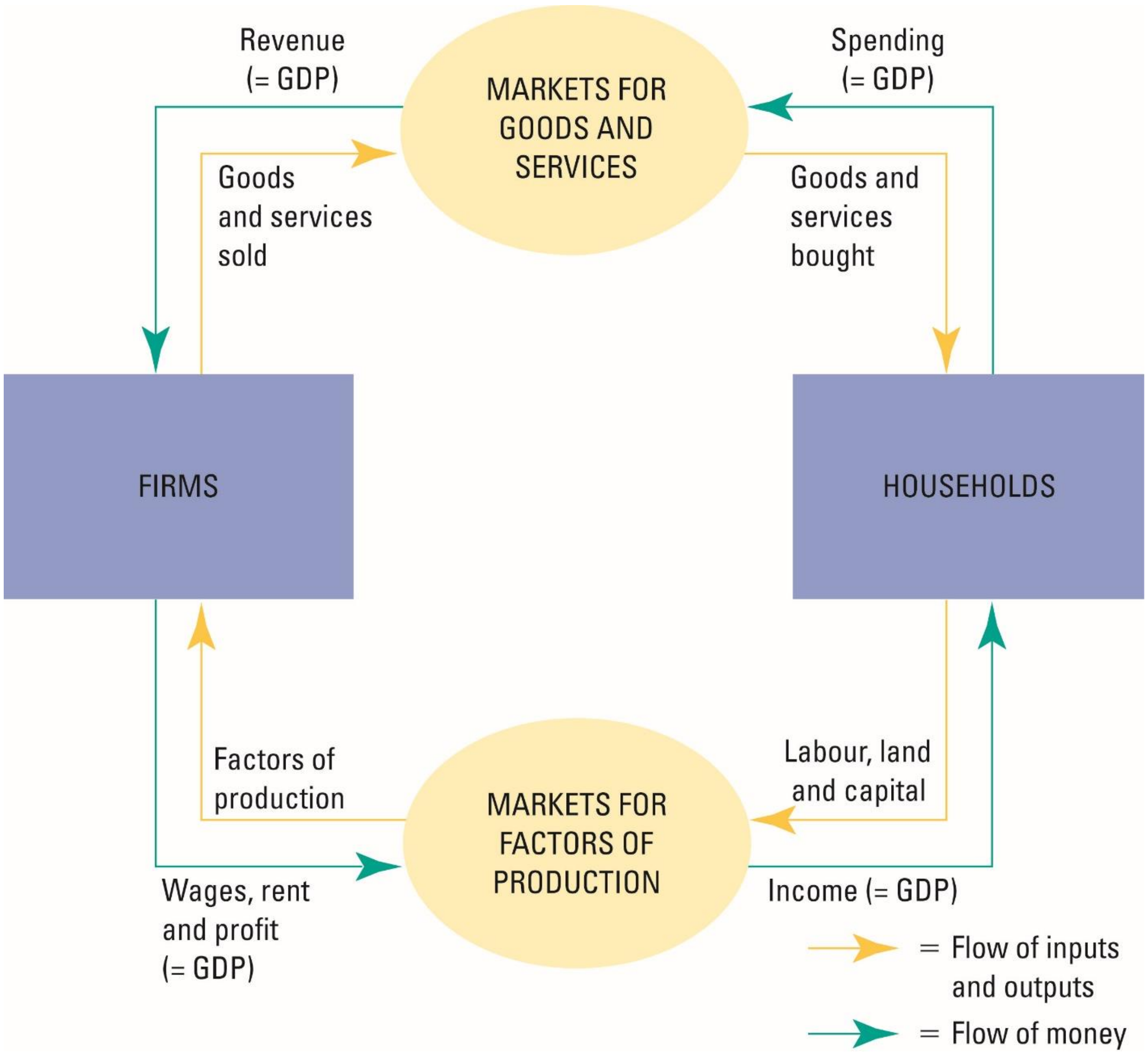
Week 1:
Measuring a nation’s income

- When judging whether the economy is doing well or poorly, it is natural to look at the total income that everyone in the economy is earning.
- For an economy as a whole, *income must equal expenditure* because:
  - Every transaction has a buyer and a seller.
  - Every dollar of spending by some buyer is a dollar of income for some seller.
  - The equality of income and expenditure can be illustrated with the circular-flow diagram.
- *Gross Domestic Product (GDP)* is a measure of the income and expenditures of an economy. There are other measures of the income of a nation (e.g., Gross National Product)
- None is perfect, and none perfectly captures living standards.



This class is about how we measure income and expenditure

Chapter Content

a) GDP definition and other measures of a nation's income

b) Expenditure approach to calculate GDP

c) Real versus nominal GDP

d) GDP per capita and well-being

e) Summary

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a. GDP definition and other measures of a nation's income

Gross Domestic Product is the total market value of all final goods and services produced within a country in a given period of time.

- "GDP is the Market Value . . ."
  - Output is valued at market prices.
- ". . . Of All Final . . ."
  - It records only the value of final goods, not intermediate goods (the value is counted only once).
- ". . . Goods and Services . . . "
  - It includes both tangible goods (food, clothing, cars) and intangible services (haircuts, house cleaning, doctor visits).

- “. . . Produced . . .”
  – It includes goods and services currently produced, not transactions involving goods produced in the past.
- “ . . . Within a Country . . .”
  – It measures the value of production within the geographic confines of a country.
- “. . . In a Given Period of Time.”
  – It measures the value of production that takes place within a specific interval of time, usually a year or a quarter (three months).

Other measures of a nation's income:

GNP or Gross National Product : adds to GDP the Net Factor Income Abroad (NFIA or SBRF for the acronym in French)

NFIA= goods and services (income) produced by Swiss residents in other countries – goods and services (income) produced by foreign residents in Switzerland

$$GNP = GDP + NFIA$$

- GNP = goods and services produced by Swiss residents in Switzerland + goods and services produced by Swiss residents in other countries
- GDP = goods and services produced by Swiss residents in Switzerland + goods and services produced by foreign residents in Switzerland

➔ Thus the net revenue of Firmenich (Swiss corporation) generated by its plant in South Africa enters Switzerland's GNP, and South Africa's GDP

Other measures:

NDP (Net Domestic Product): is GDP minus **capital depreciation**.

To account for capital consumed over the year in the form of housing, vehicle, or machinery deterioration.

Capital depreciation represents the amount of capital that would be needed to replace those depreciated assets.

NNP (Net National Product) : corrects for both capital depreciation and net factor income abroad:

NNP = GDP + NFIA – Capital Depreciation

National Income (NI) or Net National Product at factor prices: is NNP but measured at factor prices rather than market prices. We need to add up production subsidies and subtract indirect taxes (VAT for example)

NI = NNP + Subsidies – Taxes

NI = GDP + NFIA – Capital Depreciation + Subsidies - Taxes

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b. Expenditure approach to calculate GDP

Three approaches to calculating GDP

1. The Production or output approach (just seen)

The market value of all final goods and services produced within a country's borders during one year (or one quarter). Drawback: the difficulty to differentiate between intermediate and final goods. But this is equivalent to adding the value added of **all** sectors.

2. The Income approach (or GDI)

The total income earned by the factors of production owned by residents and non-residents working within the country's borders. Uses data from employment and earnings surveys. Drawback: some earnings are difficult to observe.

3. The Expenditure approach

The most widely used approach (historically) suggests simply looking at how much do households, government, non-profit institutions, and financial institutions consume within a country.

Theoretically, all three approaches must give the same value

Measuring GDP with Expenditure

GDP = Production = Consumption + Investment + Government expenditure
+ Trade Balance

$Y \equiv C + I + G + X - M$ → Trade balance (net consumption of domestically produced goods and services by non-residents)

It's an identity!!! Domestic production is identical to the consumption by residents plus the net consumption of non-residents of goods and services that were domestically produced.

C includes durable (cars, refrigerator) and non-durables goods (groceries), and services (medical, lawyers, hairdressers, etc..)

I includes business and residential investment, as well changes in inventories

G includes only government expenditure on goods and services. It does not include social transfers (unemployment, social security, etc) or interests on the debt as these payments are not done in exchange for a good or services produced during the year.

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c. Real versus nominal GDP

- *Nominal GDP* values the production of goods and services at *current prices*.
- *Real GDP* values the production of goods and services at *constant prices*.

An accurate view of the economy requires adjusting nominal to real GDP by using the GDP deflator.

The idea is to measure what has been the evolution of « real » production (goods and services) and not variations due to changes in prices, but « real » quantities remained unchanged (or moved in opposite direction).

Example

2-good economy: Hot dog and burgers

Prices and Quantities

Year	Price of hot dogs	Quantity of hot dogs	Price of hamburgers	Quantity of hamburgers
2003	€1	100	€2	50
2004	2	150	3	100
2005	3	200	4	150

Year	Calculating nominal GDP
2003	(€1 per hot dog × 100 hot dogs) + (€2 per hamburger × 50 hamburgers) = €200
2004	(€2 per hot dog × 150 hot dogs) + (€3 per hamburger × 100 hamburgers) = €600
2005	(€3 per hot dog × 200 hot dogs) + (€4 per hamburger × 150 hamburgers) = €1,200

Year	Calculating real GDP (base year 2003)
2003	(€1 per hot dog × 100 hot dogs) + (€2 per hamburger × 50 hamburgers) = €200
2004	(€1 per hot dog × 150 hot dogs) + (€2 per hamburger × 100 hamburgers) = €350
2005	(€1 per hot dog × 200 hot dogs) + (€2 per hamburger × 150 hamburgers) = €500

The « real » increase in GDP or the variation in real GDP is much less spectacular than the increase in nominal GDP would have let us believe, because there has been an important increase in prices during the period.

The evolution of nominal GDP can be very misleading if prices are rapidly changing

Real GDP can only change if quantities change. Nominal GDP changes when prices and/or quantities change.

- The *GDP deflator* is a measure of the price level calculated as the ratio of nominal GDP to real GDP times 100.

  GDP deflator = nominal GDP / real GDP * 100

- It tells us the rise in nominal GDP that is attributable to a rise in prices rather than a rise in the quantities produced.
- The GDP deflator in 2005 with base 100 in 2003 is calculated as

  = 1200/500*100 = 240

➔Between 2003 and 2005 prices went up by 140% in our 2-good economy…

- You can also convert nominal GDP into Real GDP if you have the GDP deflator (more on how to build this in the next class)

$$\text{Real GDP}_{20XX} = \frac{\text{Nominal GDP}_{20XX}}{\text{GDP deflator}_{20XX}} \times 100$$

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d. GDP per capita and well-being

- GDP per capita is the mean income produced within an economy.
- GDP per capita is the best **single** measure of economic well-being of a society.
- Higher GDP per person suggests a higher standard of living.
- But GDP per capita is still a flawed measure of living standards or quality of life.
- Income is only a small part of individual well-being even if it correlates well with other determinants (health, education, inequality, happiness)…
- The United Nations suggested the construction of a Human Development Index to capture other aspects of individual well-being.

UN’s HDI

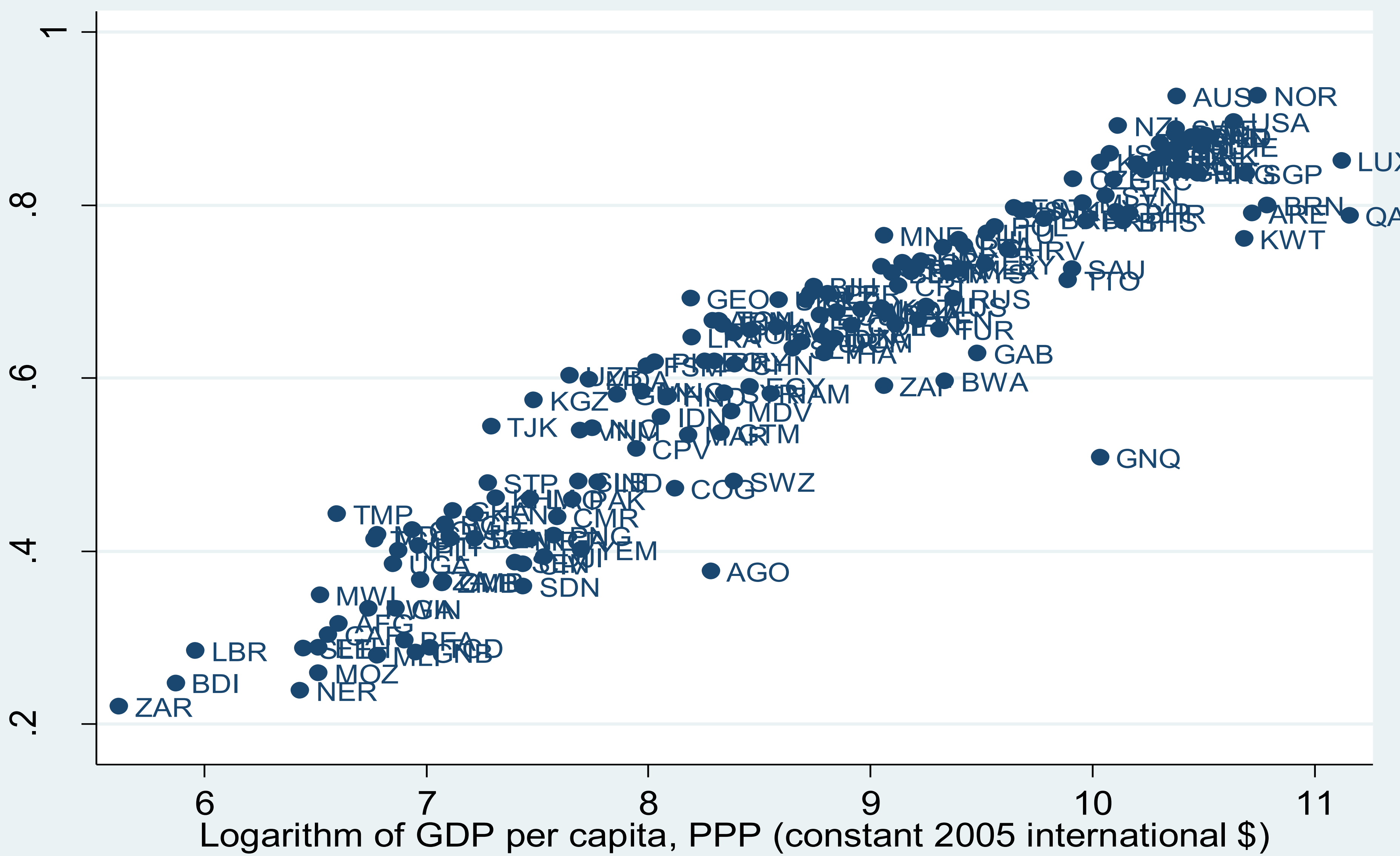
	GDP per capita	Share of 40% poorest	Life expect.	Infant mortality	Access to water	Literacy rate
Sri Lanka	2,990	22	72	18	60	89
Guatemala	3,350	8	65	48	62	54
Pakistan	2,170	21	62	91	50	36

- HDI = 1/3 Life expectancy index + 1/3 Education achievement index
  - + 1/3 GDP per capita index

But is that all that counts for well-being? Why a weight of 1/3?

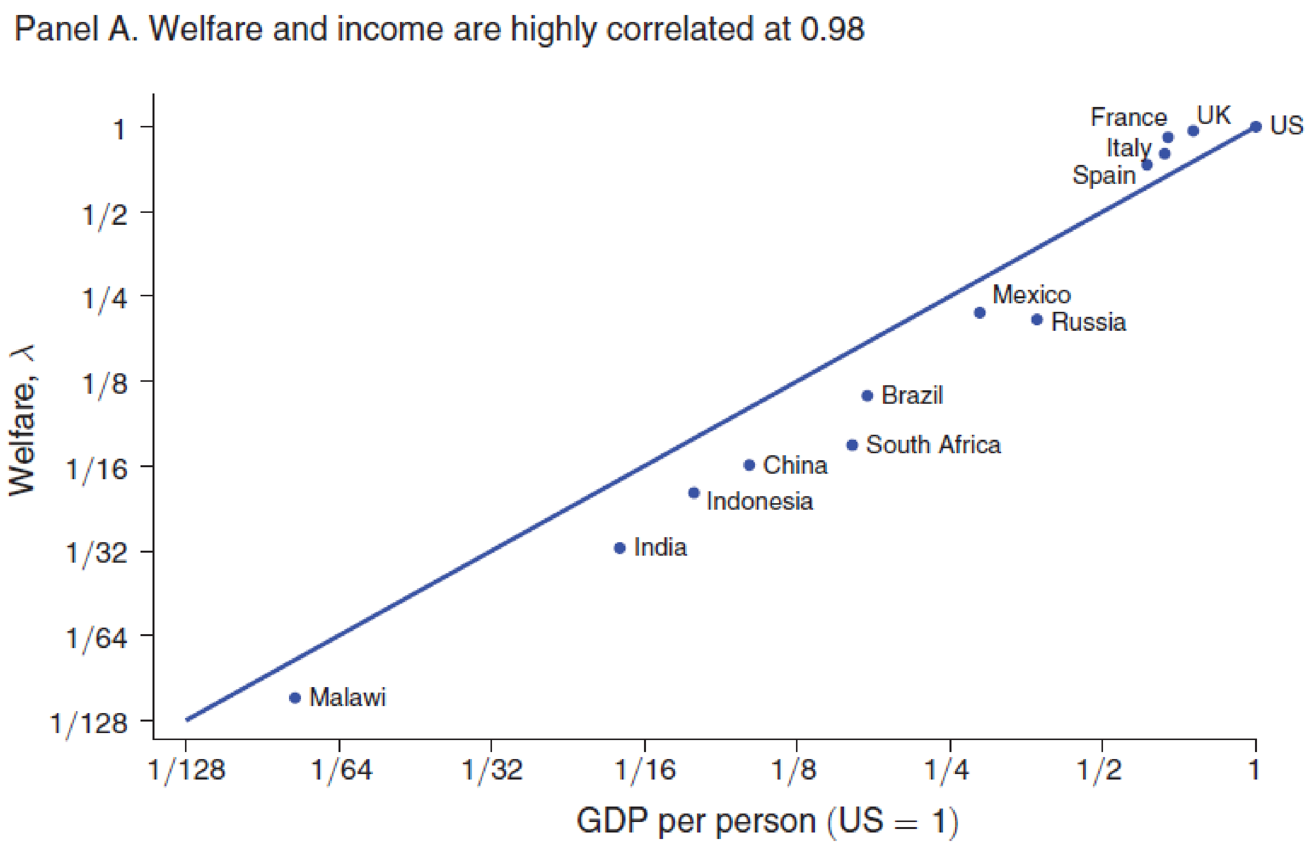
GDP per capita and human development index

GDP per capita and Human Development Index (HDI)

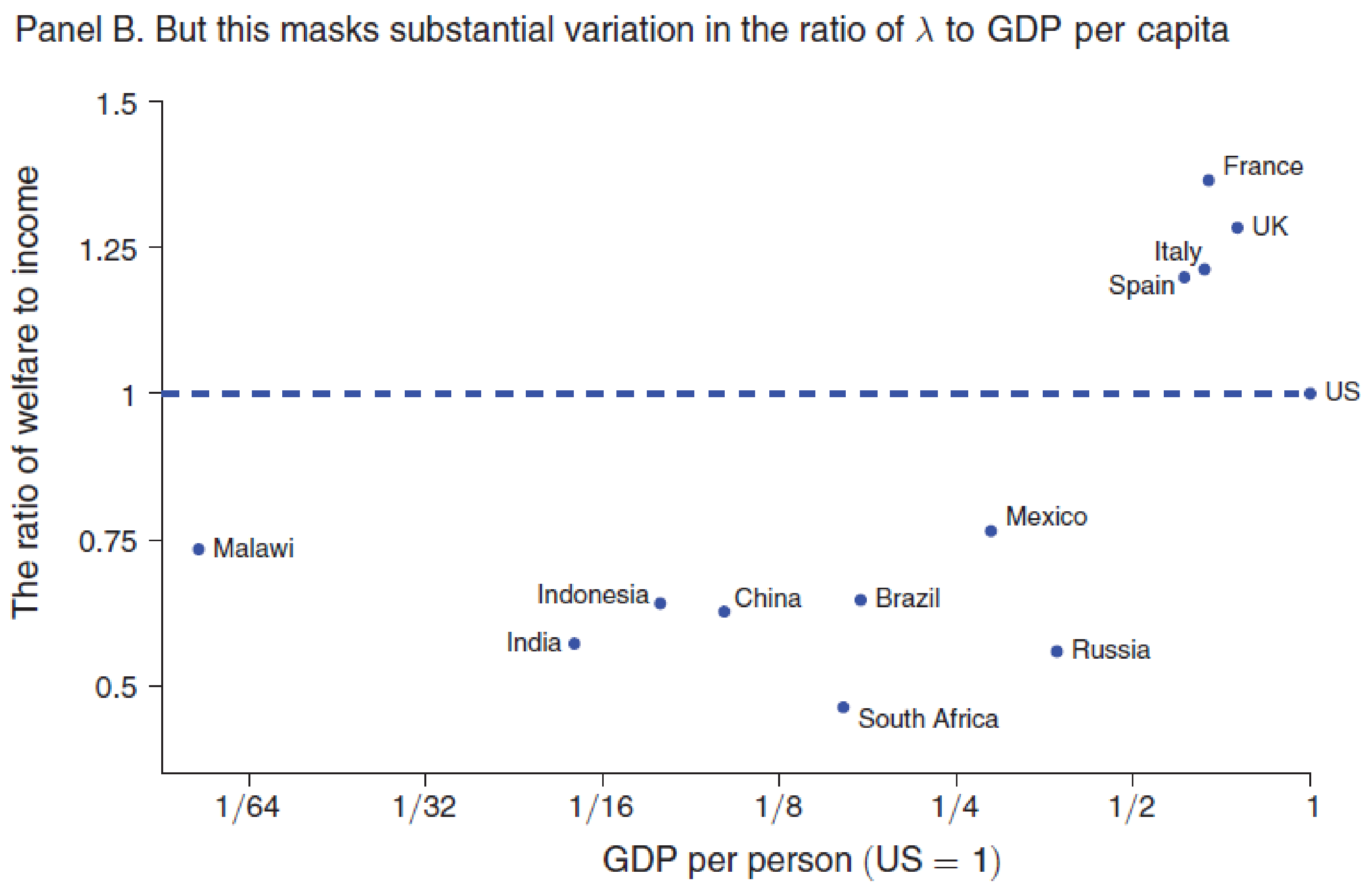


Source: World Development Indicators and Human Development Report Office, average for 2000-10.

- To solve the aggregation problem of the HDI, Jones and Klenow (AER 2016) used microeconomics tools to compute consumption equivalents for leisure, inequality, and mortality using micro data.
- They then compute a measure of relative well-being across countries using the notion of expected utility.
- Theoretically well grounded, and positively correlated with GDP per capita



- But differences between GDP per capita and Jones and Klenow (2016) welfare measure can be important



Other factors that GDP, HDI and Jones and Klenow (2016) don't consider:

- It doesn't count unpaid labor, especially housework or self consumption.
- It doesn't count illegal economic activity (smuggling or undeclared workers for example)
- It doesn't count externalities - costs that are externalized (passed on) to someone else. For example, polluting factory are able to pass on the cost of damage to health and the environment. China has been working on a measure of Green GDP.
- It includes negative goods - the clean up of a toxic waste dump is counted as a plus in terms of GDP. It also counts defensive spending: fuel expenditures in traffic jams
- It ignores distribution, quality of life, equality, social and environmental sustainability
- **The problem** is that as long as we measure the quality of life with GDP, politicians will target GDP growth rather than other aspects of well-being. It creates incentives to increase GDP, no matter what happens to environnemental degradation, income inequality, time for leisure, etc..

Further references on how to improve measures of living standards

- See the video of Joe Stiglitz on problems with the use of GDP as a measure of well-being
- See also the Swiss MONET system by the FSO to develop sustainable development indicators.
- A similar effort to measure natural capital by the UK statistical office

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e. Summary

- Because every transaction has a buyer and a seller, the total expenditure in the economy must equal the total income in the economy.
- Gross Domestic Product (GDP) measures an economy's total expenditure on newly produced goods and services and the total income earned from the production of these goods and services.
- There are other definitions of a national's income NDP, NNP, GNP, NNP, NNI, etc, which account for capital depreciation, production by non-residents, impact of government taxes and subsidies on prices, etc…
- GDP is the market value of all final goods and services produced within a country in a given period of time.
- GDP is divided among four components of expenditure: consumption, investment, government purchases, and net exports.
- Nominal GDP uses current prices to value the economy's production. Real GDP uses constant base-year prices to value the economy's production of goods and services.

- The GDP deflator—calculated from the ratio of nominal to real GDP—measures the evolution in the level of prices of goods produced within the economy.
- GDP is a relatively good measure of economic well-being because people prefer higher to lower incomes, and correlated well with other measures of well-being (health, education, leisure, etc…)
- It remains however a flawed measure of well-being because some things, such as leisure time, income inequality and a clean environment, are not captured by GDP.